Unit 13, Lesson 7: Representing Real-World Contexts with Graphs

Benchmark

MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Learning Target

- ☐ I can choose an appropriate graphical representation for a real-world context.

13.7.1 Warm-Up: Bell Work

To choose an appropriate graphical representation for data, the first step is to determine if the data is ***numerical*** (data points are numbers) or ***categorical*** (data points fall within categories).

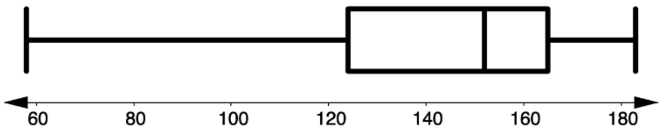
1. Complete the table. The first two rows have been completed.

Possible Question	Possible Responses	Numerical or Categorical
What is your favorite color?	red, orange, yellow, green	✓ categorical ○ numerical
How many people are in your family?	2, 3, 4, 5…	○ categorical ✓ numerical
How old are you?	11, 12, 13…	○ categorical ○ numerical
How tall are you?		○ categorical ○ numerical
What is your favorite sport?		○ categorical ○ numerical
	watermelon, apple, pear, orange	○ categorical ○ numerical

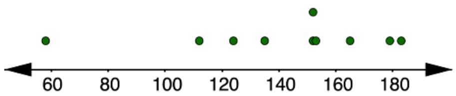
13.7.2 Exploration: Choosing the Best Graphical Representation for Numerical Data

Four graphical representations that can be used to display ***numerical data*** are box plots, histograms, line plots, and stem-and-leaf plots. Each graphical representation is shown in the table.

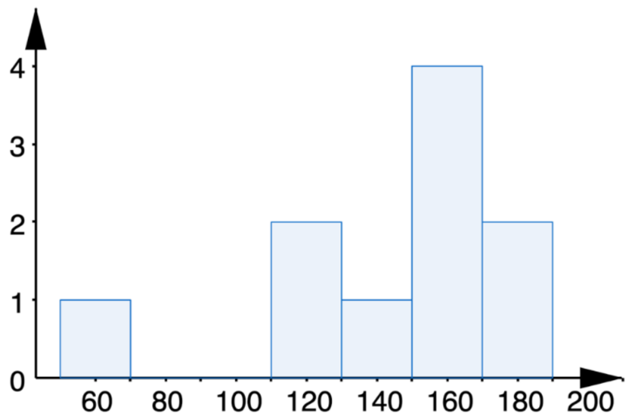
Box Plot



Line Plot



Histogram



Stem-and-Leaf Plot

Stem	Leaf
5	8
6	
7	
8	
9	
10	
11	2
12	4
13	5
14	
15	2, 2, 3
16	5
17	9
18	3

Key: 5|8 = 58

Work with your partner to complete the following.

1. Camilla wants to create a graphical display that represents the typing speeds of the 425 students at her school.

   a. Which representation(s) will best display the data?

   - ☐ box plot
   - ☐ line plot
   - ☐ histogram
   - ☐ stem-and-leaf plot

   b. Explain your reasoning.

   c. Camilla says, “Making a line plot of this data would take a lot of time and it would be hard to make sure each dot is in the right location on the number line, since speed can be given in fractions of seconds. I’d have to make 425 dots to represent each student!” Explain whether you agree with Camilla that a line plot would not best display the data.

   d. Which other display should not be considered based on the size of the data set?

2. Dallas records the highest temperature from each month over the past year in Immokalee, Florida.

$$\{76, 78, 81, 85, 90, 92, 92, 93, 91, 87, 82, 78\}$$

   a. Which representation(s) will best display the data?

   - ☐ box plot
   - ☐ line plot
   - ☐ histogram
   - ☐ stem-and-leaf plot

b. Explain your reasoning.

c. Dallas decides to change his data set to the average **weekly** high temperature. Explain whether this changes the display he should use.

3. Basil recorded the weights, rounded to the nearest pound, of 15 dogs at a pet show.

$$7, 10, 12, 22, 28, 59, 62, 65, 72, 79, 85, 98, 100, 150, 163$$

a. Explain which display would be better for the data, a line plot or a stem-and-leaf plot.

b. Basil also recorded the weights, rounded to the nearest pound, of 15 cats at the pet show.

$$4, 4, 5, 7, 7, 8, 8, 8, 9, 10, 11, 12, 13, 13, 14$$

Explain which display would be better for the data, a line plot or a histogram.

c. Basil updated his data to show the weights of the cats to the nearest tenth.

$$4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6$$

Explain which display would be better for the data, a line plot or a histogram. If your recommendation is different from Part B, explain why.

4. Ms. Rallen asked her students how many pets they each have, and wants to use a graphical representation that will allow her students to see the overall shape of the data and give them enough information to calculate the 5-number summary.

   a. Which graphical representations should Ms. Rallen consider?

   - ☐ box plot
   - ☐ line plot
   - ☐ histogram
   - ☐ stem-and-leaf plot

   b. Explain your reasoning.

13.7.3 Bring It Together: Choosing the Best Graphical Representation for Numerical Data

Line plots are best for smaller sets of data and when the goal is to see each value over a scale, making it easier to calculate measures of center and spread.

Histograms are best for larger sets of data, data over a large range, or continuous data. One disadvantage is that the display cannot be used to calculate measures of center and spread since individual values cannot be determined from the display.

Stem-and-leaf plots are helpful in visualizing the shape of a distribution, while still being able to see the individual values.

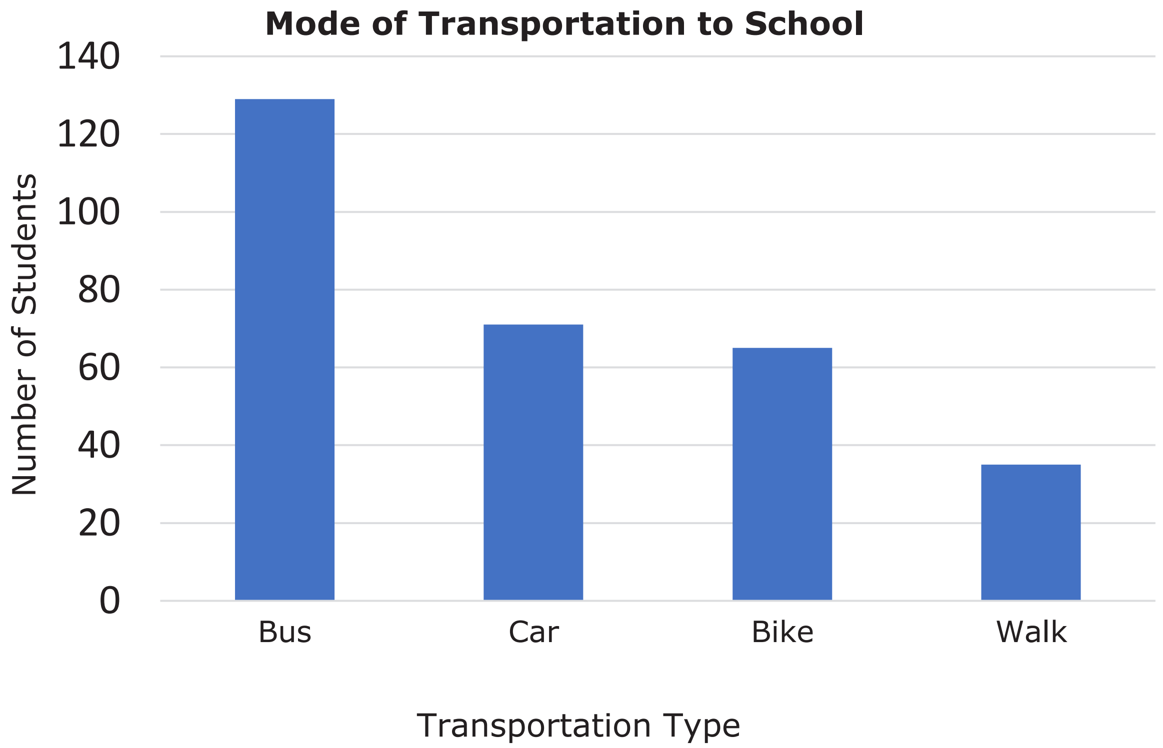
Box plots are helpful in visualizing the shape of a distribution and quickly identifying the five-number summary. This display cannot be used to find the mean.

13.7.4 Guided Instruction: Graphical Representations for Categorical Data

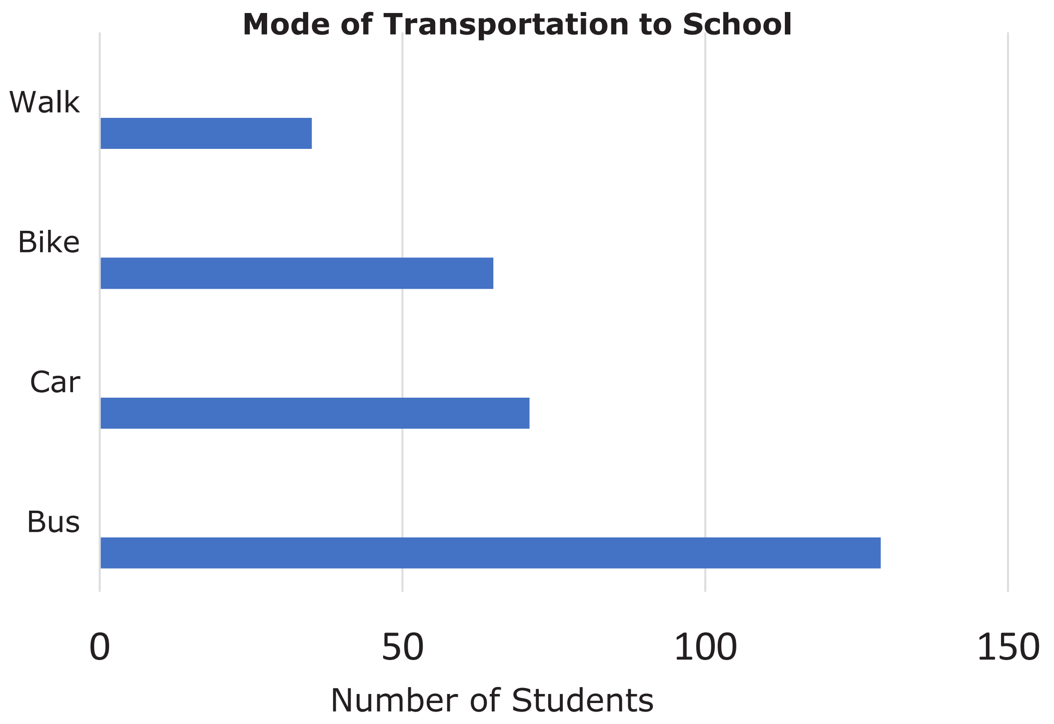
Two graphical representations that can be used to display categorical data are bar graphs and circle graphs.

A ***bar graph*** is a visual display of categorical data values where each category is represented by a bar whose height represents the number in that category. Bar graphs can be represented vertically or horizontally.

The vertical bar graph shows data collected from a survey on how students get to school.

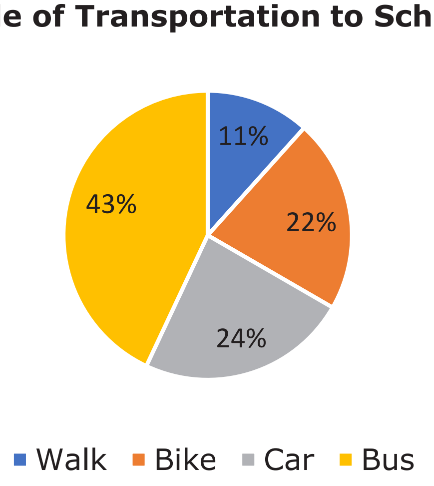


The same data is also displayed in a horizontal bar graph.



A ***circle graph*** is a visual display of categorical data. The whole set of data is represented by the circle and its interior. The categories are represented by fractional parts of the circle. This is also called a pie chart.

The circle graph below shows the same data.



1. Complete the statements.

The bar graph visually shows

- ○ how each part relates to the whole.
- ○ how each part compares to the other parts.

The circle graph shows

- ○ how each part relates to the whole.
- ○ how each part compares to the other parts.

2. Greer wants to convince the head chef at his school to include oat milk as an option at meals, so he asks a random sample of students from his school what their favorite kind of milk is: dairy, oat, or soy. He wants to use a graphical representation that makes it easy to compare the number of students who prefer oat milk to the total number of students surveyed.

Jordan asks a random sample of students from his middle school which of these apps is their favorite: TikTok, Snapchat, Instagram, Pinterest, YouTube, WhatsApp, Kik, or Discord. He wants to display the data so it is easy to compare the different categories to each other.

Complete each statement.

- ○ Greer
- ○ Jordan

should use a bar graph. A bar graph is

the best option for this student because it

- ○ visually shows how each part relates to the whole.
- ○ visually shows how each part compares to the other parts.

- ○ Greer
- ○ Jordan

should use a circle graph. A circle graph

is the best option for this student because it

- ○ visually shows how each part relates to the whole.
- ○ visually shows how each part compares to the other parts.

13.7.5 Your Turn!

1. Mr. Shapiro's seventh grade class read more books than any other class in the middle school over the summer. The number of books read by the students in his class is shown below.

$$\{1, 1, 3, 5, 5, 6, 6, 6, 6, 7, 8, 9, 10, 10, 11, 12, 12, 14, 14, 15, 15, 15, 17, 17, 20, 23, 25, 27\}$$

He wants to make a graph that shows the number of books read by each student. Which graphical representation would you recommend for this situation? Explain your reasoning.

2. Kendrick wants to make a graph that will show what proportion of students in his homeroom like science class best when compared to language arts, math, and social studies. Explain which graphical representation would best show this comparison.

13.7.6 Wrap-Up: Growth Mindset

1. Every mistake a person makes, their brain actually grows! Take a moment to celebrate your mistakes by summarizing your favorite mistake from today.

   My favorite mistake from today was …